

Uveal melanoma: Treatment approaches and prognostic outcomes based on the American joint committee on cancer staging

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ABSTRACT

Purpose: To evaluate the clinical characteristics, treatment outcomes, and prognosis of uveal melanoma (UM) based on the 8th edition of the American Joint Committee on Cancer (AJCC) staging system.

Design: Retrospective study.

Methods: Data on the clinical presentation, diagnostic methods, treatment approaches, and long-term outcomes of patients diagnosed with UM at a tertiary referral center in Thailand between 2010 and 2024 were included and analyzed.

Results: Of 106 patients included, the mean (range) age at diagnosis was 51.6 (16–86) years. The most common tumor location was the choroid (n = 100, 94.3%), followed by the ciliary body (n = 4, 3.8%) and iris (n = 2, 1.9%). AJCC tumor staging was T1 in 18.9%, T2 in 17.9%, T3 in 48.1%, and T4 in 15.1% of cases. Primary treatment included enucleation (n = 57, 53.8%), Ruthenium-106 (¹⁰⁶Ru) brachytherapy (n = 31, 29.2%), endoresection combined with ¹⁰⁶Ru brachytherapy (n = 17, 16.0%), and exoresection (n = 1, 0.9%). At a median follow-up of 37.3 months, the overall globe salvage rate was 44.3%. Kaplan-Meier analysis showed a 5-year and 10-year distant metastasis rate of 22%, with disease-specific survival rates of 82% and 75%, respectively. The incidence of melanoma-related death was 3.4 per 100 person-years, increasing with tumor stage. In multivariate analysis, distant metastasis was the only significant factor associated with the overall survival outcome.

Conclusions: This study provides insights into the clinical features, treatment outcomes, and prognostic factors of UM in a Southeast Asian cohort. Tumor stage and the presence of metastasis significantly influence prognosis, highlighting the importance of early detection and individualized treatment strategies to improve patient outcomes.

1. Introduction

Uveal melanoma (UM), originating from melanocytes, is the most common primary intraocular malignancy. The uveal tract is the second most common site of melanoma after the skin, with most tumors located in the choroid, followed by the ciliary body and iris.^{1,2}

The incidence of UM varies significantly across regions and is influenced by demographic and environmental factors, including Caucasian ethnicity, light-colored irises, fair skin and hair, a family

history of UM, and specific genetic predispositions.^{3,4} A review of UM in Asian populations reported an incidence of 0.2–0.6 cases per million, considerably lower than the 5–6 cases per million observed in Caucasians.⁵ Moreover, UM in Asian patients typically presents at a younger age, between 40 and 55 years, compared to the mean presentation age of 58 years in Caucasians.⁵ Retrospective case series have shown that although the relative risk of metastasis in Asian patients is slightly elevated, the rates of melanoma-related metastasis and mortality are comparable across different racial groups.⁶

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Despite these findings, research on UM in the Southeast Asian region remains scarce.^{7,8} As a primary referral center for intraocular tumors in Thailand and neighboring countries, this study aims to investigate the presenting features, clinical course, treatment approaches, prognosis, and outcomes of UM patients in this population.

2. Methods

We conducted a retrospective review of medical records for all patients diagnosed with UM and treated in the Department of Ophthalmology at Ramathibodi Hospital, Mahidol University, Bangkok, Thailand, between January 2010 and April 2024. Institutional Ethics Committee (MURA2025/802) approval was obtained and informed consent was waived due to the retrospective nature of the study. The study adhered to the tenets of the Declaration of Helsinki.

The patient's record data were reviewed for demographic information, clinical features, management, outcomes, and survival rates. The demographic data included gender, ethnicity, age of presentation, presenting symptoms and duration, laterality, visual acuity, and underlying diseases. Clinical features of tumor included tumor origin (iris, ciliary body, choroid), extraocular extension, pigmentation (melanotic, amelanotic, mixed), subretinal fluid, lipofuscin, greatest tumor thickness, largest basal diameter, anteroposterior location of tumor epicenter (macula, macula to equator, anterior to equator, iridociliary involvement), quadrantic location of tumor epicenter (macula, superior, temporal, nasal, inferior), posterior margin to the optic disc, and posterior margin to foveola. All patients underwent a complete ocular examination, and tumor dimensions were measured by ultrasonography. Tumor non-regression was defined clinically by the persistence or recurrence of activity markers, specifically the presence of subretinal fluid and orange pigment (lipofuscin) overlying the lesion, and/or tumor growth during follow-up examinations. Additional multimodal imaging was performed including color fundus photography, ultrasonography, spectral-domain optical coherence tomography (SD-OCT), and fundus autofluorescence (FAF), fluorescein angiography (FFA), and indocyanine green angiography (ICG).

The classification of UM by size of the tumor is based on the criteria published by the Collaborative Ocular Melanoma Study (COMS). Small tumors had a maximum thickness of 2.5 mm and a maximum base of 16 mm; medium-sized tumors had a thickness between 2.5 and 10 mm and a maximum base of 16 mm; and for large tumors, the maximum thickness was higher than 10 mm or maximum base more than 16 mm.⁹ The other classification based on the 8th edition American Joint Committee (AJCC) for posterior uveal melanoma, each tumor was categorized into T1, T2, T3, and T4 by the extent of the primary tumor (T), the presence or absence of lymph node involvement (N), and the presence or absence of distant metastasis (M).¹⁰

All patients were evaluated and managed by a single ocular oncologist (D.R.). Treatment selection was based on the tumor's size, staging, and the patient's preference. The primary treatment included Ruthenium-106 plaque brachytherapy (¹⁰⁶Ru), pars plana vitrectomy endoresection combined with ¹⁰⁶Ru, exoresection, and enucleation. All enucleated globes were revealed for histopathological results by ocular pathologists. Systemic evaluation for systemic metastasis was performed before the treatment. Long-term surveillance with medical oncologists includes physical examination, liver function test, and imaging of chest and abdomen. The outcome after the treatment regarding visual acuity, globe salvage, local tumor recurrence, systemic metastasis, and survival status was documented.

For ¹⁰⁶Ru plaque brachytherapy, our institution utilizes two models of plaques selected according to the size and location of the tumor. The first model is the round-type CCA plaque with a diameter of 15 mm, and the second is the COB notch-type plaque with a diameter of 20 mm. The appropriate plaque was chosen based on the tumor's dimensions and position. A scheduled radiation dose of 80–100 Grays (Gy) was delivered to the tumor apex. Treatment durations were calculated using Plaque

Simulator software (BEBIG, 2008).¹¹

3. Statistical analysis

The statistical analysis was performed using STATA v17 (Stata-Corp 2021, College Station, TX, USA). Continuous variables were checked for normality of distribution by Kolmogorov-Smirnov test, and were described using means, medians, and ranges. The ANOVA test and Kruskal-Wallis's test were used to compare parametric continuous variables and nonparametric continuous data, respectively. Categorical variables were compared using the Chi-square test. Bonferroni-adjusted significance tests were used for pairwise comparisons.

Time to metastasis was measured from the date of the first presentation until the date of metastasis. Univariate and multivariate logistic regression analyses were performed to assess the risk factors for distant metastasis and mortality and evaluate the hazard ratio (HR) with a 95% confidence interval (CI). The rate of distant metastasis and survival at 1, 3, 5, and 10 years were calculated using the Kaplan-Meier analysis. The Cox proportional hazards regression model was used for cause-specific hazards for competing risk endpoints. A p-value of < 0.05 was considered statistically significant.

4. Results

In this single-center 15-year study, 106 patients with UM were included. The tumor location was choroid in 100 patients (94.3%), ciliary body in 4 patients (3.8%) and iris in 2 patient (1.9%). Demographic and clinical characteristics are summarized in Table 1. No gender predilection was observed. The patient's ethnicity was Asian (100 cases, 94.3%; 96 Thais, 2 Cambodians, 1 Myanmar, and 1 Chinese), and Caucasian (6 cases, 5.7%; 5 Europeans and 1 American). The mean age at diagnosis was 51.6 ± 16.1 years (range, 16–86 years). The most common presenting symptom was visual loss (78.3%), followed by floaters and/or photopsia (12.3%). The median duration of symptoms prior to diagnosis was 2.5 months. The median initial visual acuity (VA) was 0.88 logMAR (mean 1.09 ± 0.94; range, 20/20 to no light perception). Three patients (2.8%) had associated oculodermal melanosis. Most patients were diagnosed based on clinical evaluation combined with multimodal imaging, including color fundus photography, ultrasonography, SD-OCT, FAF, FFA, and ICG. In 10 patients (9.4%) with diagnostic uncertainty, such as in cases of amelanotic lesions, fine-needle aspiration biopsy was required and confirmed the diagnosis of melanoma.

Staging for prognosis was classified according to the 8th edition of the AJCC. The majority of patients were classified as stage 2 (58 cases; 54.7%), followed by stage 3 (27 cases; 25.5%), stage 1 (19 cases; 17.9%), and stage 4 (2 cases; 1.9%). Tumors were further categorized based on size into T categories: T1 (20 cases; 18.9%), T2 (19 cases; 17.9%), T3 (51 cases; 48.1%), and T4 (16 cases; 15.1%). Extraocular extension was observed in 1.9% of cases (2 patients), occurring in the T3 and T4 categories; diagnosis was established clinically in one patient and confirmed histopathologically in the other. Features and tumor characteristics based on the T categories are summarized in Table 2.

There were no significant differences in patient age, gender, or duration of symptoms among the groups. However, the median initial VA differed significantly between T1 and T3 ($P = 0.006$) and between T1 and T4 ($P < 0.001$). The median VA values in logMAR were 0.1, 0.7, 1.2, and 2.3 in T1, T2, T3, and T4, respectively. Subretinal fluid overlying or surrounding the tumor was observed in 88.8% of patients and did not vary significantly across T categories.

The shape of the tumor mass showed a significant difference among T categories ($P = 0.001$). Dome-shaped configurations were most common in T1 tumors (77.8%), while the proportion of mushroom-shaped tumors increased with higher T stages, being most prominent in T4 (85.7% in T4 vs. 11.1% in T1, $P = 0.012$). Pigmentation did not differ significantly among T categories ($P = 0.635$), with melanotic masses being the most

Table 1
Demographics and Clinical Features of Patients with Uveal Melanoma.

Characteristic (n = 106)	Statistics
Age at diagnosis (years); mean ± SD (range)	51.56 ± 16.1 (16–86)
Biological sex; n (%)	
Male	51 (48.1)
Female	55 (51.9)
Ethnicity; n (%)	
Asian	100 (94.3)
Caucasian	6 (5.7)
Anatomical location; n (%)	
Choroidal melanoma	100 (94.3)
Ciliary body melanoma	4 (3.8)
Iris melanoma	2 (1.9)
Laterality; n (%)	
OD	46 (43.4)
OS	60 (56.6)
Symptoms; n (%)	
Visual loss	83 (78.3)
Floater/flashing	13 (12.3)
Multiple complaint	3 (2.8)
Visual field defect	2 (1.9)
Metamorphopsia	2 (1.9)
No symptoms (Accidental finding)	3 (2.8)
Symptom duration (months); mean (range)	2.5 (0.1–96.0)
Initial Visual Acuity (logMAR); median, mean ± SD (range)	0.88, 1.09 ± 0.94 (0–3)
Associated Oculodermal Melanosis	3 (2.8)
Extraocular Extension	2 (1.9)
Pre-treatment Distant Metastasis	2 (1.9)
Liver metastases	1 (0.9)
Multiple metastases	1 (0.9)
AJCC 8th Edition Staging	
1	19 (17.9)
2	58 (54.7)
3	27 (25.5)
4	2 (1.9)
T1	20 (18.9)
T2	19 (17.9)
T3	51 (48.1)
T4	16 (15.1)
Tumor Size (COMS Classification)	
Small (S)	3 (2.8)
Medium (M)	53 (50.0)
Large (L)	50 (47.2)

AJCC, The American Joint Committee on Cancer; COMS, Collaborative Ocular Melanoma Study; OD, Oculus Dextrus; OS, Oculus Sinister; SD, Standard Deviation.

common across all groups. Quadrant location of the tumor also did not vary significantly among T categories ($P = 0.089$). Regarding the anteroposterior tumor epicenter, the macula-to-equator region was the most common location across all T stages (T1–T4). Tumors located at the macula were observed in 15 cases, including 7 in T1, 7 in T2, 1 in T3, and none in T4 ($P = 0.001$). Other tumor epicenter locations did not show a statistically significant difference across T categories. No significant differences were observed in the posterior tumor margin distance to the optic disc ($P = 0.369$) or proximity to the foveola ($P = 0.788$). The mean tumor thickness was 9.4 ± 4.2 mm (range, 2.4–18.0 mm), and the mean largest basal diameter was 11.9 ± 4.0 mm (range, 3.0–22.0 mm).

5. Treatment modalities and outcomes

The treatment modality for uveal melanoma was determined based on tumor staging, size, thickness, location, and patient preference. The most commonly performed treatment was enucleation, undertaken in 57 patients (53.8%), primarily for T3 and T4 tumors, with a mean tumor thickness of 11.9 mm (range 5.0–18.0 mm). One case with a tumor thickness of 5 mm underwent enucleation due to its epipapillary location. ^{106}Ru brachytherapy was utilized in 31 patients (29.2%), predominantly for T1 and T2 tumors, with a mean tumor thickness of 4.7 mm (range 2.4–7.5 mm). Endoresection was combined with ^{106}Ru

brachytherapy in 17 patients (16.0%), with a mean tumor thickness of 10.0 mm (range 5.4–16.6 mm). Additionally, exoresection using the partial lamellar sclerouvectomy technique was performed in one patient (0.94%) with thick iridociliary melanoma (tumor thickness of 11.0 mm). One patient underwent secondary enucleation due to local recurrence of ciliochoroidal melanoma following primary endoresection. The primary treatment modalities based on T staging are summarized in Table 3.

Histopathology was performed in 70 patients (66.0%) and showed mixed cell type tumor (mixture of spindle and epithelioid cells) in 60 patients (85.7%), epithelioid cell-type tumor in 7 (10.0%) patients, and spindle cell-type tumor in 3 (4.3%) patients. There was no significant difference in cell type among each T category ($P = 0.558$) as shown in Table 4.

The median follow-up time was 37.3 months (range 1–130 months, excluding a patient who died postoperatively from air embolism).¹² Follow-up period was less than 1 year, 1–3 years, 3–5 years, 5–10 years, and more than 10 years in 21 (20.0%) patients, 32 (30.5%) patients, 25 (23.8%) patients, 25 (23.8%) patients, and 2 (1.9%) patients, respectively.

The outcome measures of uveal melanoma were evaluated over a median follow-up duration of 37.3 months. Globe salvage was achieved in 47 patients (44.3%), with the highest success rate observed in T1 tumors (90.0%) and the lowest in T4 tumors (12.5%). The likelihood of globe preservation decreased significantly with increasing T stage ($p < 0.001$), as shown in Table 3.

The median final VA among all eyes preserved was 1.0 logMAR (range 0–3). Stratified by treatment modality, the median VA was 0.4 logMAR for ^{106}Ru brachytherapy, 1.3 logMAR for endoresection, and 2.3 logMAR for exoresection. When classified by AJCC T category, the median VA was 0.3, 1.0, 1.3, and 1.25 logMAR for T1, T2, T3, and T4, respectively), as shown in Table 3. No significant differences were observed among treatment modalities or across T stages.

6. Systemic metastasis and survival outcomes

Post-treatment surveillance was conducted in a multidisciplinary setting. In the absence of molecular risk stratification, a uniform surveillance protocol was applied to all patients. This included physical examination and liver function tests performed at 6-month intervals. Imaging surveillance comprised liver ultrasonography and chest radiography performed every 6–12 months, or more frequently depending on clinical suspicion, and medical oncologists justification. The median follow-up duration was 37.3 months. At initial presentation, two patients (1.9%) were found to have distant metastases, one with liver metastasis and the other with multiple metastases. During the follow-up period, an additional 12 patients (11.5%) developed distant metastases (1, 3, 5, and 3 patients in T1, 2, 3, and 4, respectively) with a median time to metastasis of 2.3 years from the initial presentation.

The most frequently observed metastatic pattern was involvement of multiple organs (5 patients), followed by liver metastases (4 patients), lung metastases (2 patients), and bone metastasis (1 patient). According to the Kaplan-Meier time-to-event analysis (Table 5), the cumulative rates of distant metastasis across all T stages were 4% at 1 year, 12% at 3 years, 22% at 5 years, and 22% at 10 years. Subgroup analysis of Kaplan-Meier estimates for melanoma-related metastasis rates at 5 years revealed rates of 8% for T1, 37% for T2, 21% for T3, and 30% for T4 stages (Fig. 1).

In the Cox proportional hazards regression analysis, factors associated with an increased risk of systemic metastasis included ciliary body involvement (HR 1.46, 95% CI: 0.45–4.68, $P = 0.53$), T4 staging compared to T1 (HR 3.63, 95% CI: 0.38–34.92, $P = 0.27$), and tumor non-regression (HR 2.47, 95% CI: 0.32–18.96, $P = 0.39$). However, due to the small number of metastatic cases, no statistically significant associations were identified.

For survival outcomes, 16 patients (15.1%) died during the study period. Of these, 13 patients (12.3%) succumbed to melanoma-related

Table 2
Demographic and Clinical Features of Ocular Melanoma Patients Stratified by AJCC T category.

Characteristic	Total (n = 106)	T1 (n = 20)	T2 (n = 19)	T3 (n = 51)	T4 (n = 16)	P value
Age at diagnosis (year); mean ± SD	51.6 ± 16.1	49.0 ± 13.8	54.5 ± 17	50.9 ± 16.4	52.3 ± 16.5	0.835
Biological sex; n (%)						0.736
Male	51 (48.1)	10 (50.0)	8 (42.1)	26 (51.0)	6 (37.5)	
Female	55 (51.9)	10 (50.0)	11 (57.9)	25 (49.0)	10 (62.5)	
Duration of symptoms (months); median (range)	2.5 (0.1–96.0)	3.0 (0.2–60.0)	4.0 (0.1–12.0)	2 (0.1–36)	2.5 (0.1–96)	0.391
Median initial VA (logMAR)	0.9	0.1	0.7	1.2	2.3	0.001
Subretinal fluid; n (%)						0.587*
Present	88 (83.0)	18 (90.0)	16 (84.2)	43 (84.3)	11 (68.8)	
Absent	12 (11.3)	1 (5.0)	2 (10.5)	6 (11.8)	3 (18.0)	
Data not available	6 (5.7)	1 (2.0)	1 (2.0)	2 (3.9)	2 (3.9)	
Tumor shape; n (%)						0.001*
Dome-shaped	46 (43.4)	14 (70.0)	10 (52.6)	21 (41.2)	1 (6.2)	
Mushroom-shaped	46 (43.4)	2 (10.0)	9 (47.4)	23 (45.1)	12 (75.0)	
Plateau	1 (0.9)	1 (5.0)	0 (0.0)	0 (0.0)	0 (0.0)	
Bilobed	4 (3.8)	1 (5.0)	0 (0.0)	2 (3.9)	1 (6.2)	
Data not available	9 (8.5)	2 (10.0)	0 (0.0)	5 (9.8)	2 (3.9)	
Pigmentation; n (%)						0.635
Melanotic	84 (79.2)	18 (90.0)	14 (73.7)	39 (76.5)	13 (81.2)	
Amelanotic	9 (8.5)	1 (5.0)	4 (21.1)	3 (5.9)	1 (6.2)	
Partial pigmentation	7 (6.6)	1 (5.0)	1 (5.3)	5 (9.8)	0 (0.0)	
Data not available	6 (5.7)	0 (0.0)	0 (0.0)	4 (7.8)	2 (3.9)	
Quadratic location; n (%)						0.089*
Epipapillary	4 (3.8)	1 (5.0)	2 (10.5)	1 (2.0)	0 (0.0)	
Macula	10 (9.4)	6 (30.0)	2 (10.5)	1 (2.0)	1 (6.2)	
Superior	27 (25.5)	5 (25.0)	5 (26.3)	12 (23.5)	5 (31.2)	
Nasal	22 (20.8)	3 (15.0)	3 (15.8)	13 (25.5)	3 (18.8)	
Inferior	19 (17.9)	3 (15.0)	5 (26.3)	10 (19.6)	1 (6.2)	
Temporal	22 (20.8)	2 (10.0)	2 (10.5)	12 (23.5)	6 (37.5)	
Data not available	2 (1.9)	0 (0.0)	0 (0.0)	2 (3.9)	0 (0.0)	
Anteroposterior tumor epicenter; n (%)						0.001*
Macula	15 (14.2)	7 (35.0)	7 (36.8)	1 (2.0)	0 (0.0)	
Macula to equator	60 (56.6)	10 (50.0)	8 (42.1)	33 (64.7)	9 (56.2)	
Anterior to equator	24 (22.6)	2 (10.0)	4 (21.1)	13 (25.5)	5 (31.2)	
Iris-ciliary body	5 (4.7)	1 (5.0)	0 (0.0)	3 (5.9)	1 (6.2)	
Data not available	2 (1.9)	0 (0.0)	0 (0.0)	1 (2.0)	1 (6.2)	
Posterior tumor margin; n (%)						0.369*
≤ 3 mm from optic disc	52 (49.1)	7 (35.0)	10 (52.6)	27 (52.9)	8 (50.0)	
> 3 mm from optic disc	41 (38.7)	11 (55.0)	9 (47.4)	16 (31.4)	5 (31.2)	
Data not available	12 (11.3)	2 (10.0)	0 (0.0)	8 (15.7)	3 (18.8)	
Proximity to foveola (mm); median (range)	3 (0–20)	2 (0–10)	3.7 (0–13)	3 (0–20)	3.0 (0–9)	0.788
Tumor thickness (mm); mean ± SD	9.4 ± 4.2	4.0 ± 1.0	6.3 ± 1.5	10.8 ± 2.5	15 ± 2.3	< 0.001
Largest basal diameter (mm); mean ± SD	11.9 ± 4.0	7.4 ± 2.0	10.2 ± 1.8	13.1 ± 2.5	16.3 ± 3.0	< 0.001

AJCC, The American Joint Committee on Cancer; SD, standard deviation; VA, visual acuity.

* P value excluded non-available data.

Table 3
Primary Treatment and Treatment Outcomes of Ocular Melanoma Patients Stratified by AJCC T category.

	Total (n = 106)	T1 (n = 20)	T2 (n = 19)	T3 (n = 51)	T4 (n = 16)
Primary treatment; n (%)					
¹⁰⁶ Ru brachytherapy	31 (29.2)	17 (85.0)	10 (52.6)	3 (5.88)	1 (6.25)
Exoresection	1 (0.94)	0 (0)	0 (0)	1 (1.96)	0 (0)
Endoresection	17 (16.04)	1 (5.0)	3 (15.79)	11 (21.57)	2 (12.50)
Primary enucleation	57 (53.8)	2 (10.0)	6 (31.58)	36 (70.59)	13 (81.25)
Treatment outcomes; n (%)					
Tumor regression following primary treatment (n = 49)	43 (87.7)	16 (88.9)	12 (92.3)	13 (92.9)	2 (50)
Median final VA (logMAR) (n = 47)*	1.0	0.3	1.0	1.3	1.25
Globe preservation (n = 106)	47 (44.3)	18 (90)	13 (68.4)	14 (27.5)	2 (12.5)

PDT, Photodynamic Therapy; ¹⁰⁶Ru, Ruthenium-106; TTT, Transpupillary Thermotherapy; VA, Visual Acuity.

* After excluding 57 cases of primary enucleation and 2 cases of secondary enucleation.

metastasis, while the remaining 3 deaths were due to non-melanoma-related causes. According to Kaplan-Meier analysis, the overall survival rates at 1, 3, 5, and 10 years were 98%, 88%, 78%, and 71%, respectively, whereas the disease-specific survival rates for the same time intervals were 100%, 92%, 82%, and 75%, respectively. Survival

rates stratified by T category are presented in Figs. 2 and 3. The incidence rate of overall mortality in this study was 4.2 cases per 100 person-years. Mortality rates based on the T category were 1.4, 3.5, 4.1, and 9.6 cases per 100 person-years, showing an increasing trend with higher T stages from T1 to T4. Furthermore, in the Cox proportional

Table 4
Pathological Cell Types of Ocular Melanoma Patients Stratified by AJCC T category.

Pathological cell type; n (%)	Total (n = 70)	T1 (n = 3)	T2 (n = 9)	T3 (n = 43)	T4 (n = 15)	P value
Spindle	3 (4.3)	0 (0.0)	0 (0.0)	3 (7.0)	0 (0.0)	0.708
Mixed	60 (85.7)	3 (100.0)	8 (88.9)	37 (86.0)	12 (80.0)	
Epithelioid	7 (10.0)	0 (0.0)	1 (11.1)	3 (7.0)	3 (20.0)	

Table 5
Kaplan-Meier Analysis of Distant Metastasis and Survival Outcome of Ocular Melanoma Patients Stratified by AJCC T category.

KM analysis	Total (n = 106)	T1 (n = 20)	T2 (n = 19)	T3 (n = 51)	T4 (n = 16)
Systemic metastasis; mean % ± SD					
1 year	4.3 ± 2.1	0	0	6.7 ± 3.7	7.7 ± 7.4
3 years	11.6 ± 3.7	7.7 ± 7.4	21.3 ± 13.4	9.3 ± 4.5	16.1 ± 10.5
5 years	22.2 ± 5.6	7.7 ± 7.4	37.0 ± 17.7	21.0 ± 7.5	30.1 ± 15.5
10 years	22.2 ± 5.6	-	-	21.0 ± 7.5	-
Overall survival; mean % ± SD					
1 year	97.9 ± 1.5	100.0	100.0	95.6 ± 3.1	100.0
3 years	88.2 ± 3.7	90.9 ± 8.7	100.0	90.2 ± 4.7	68.8 ± 13.3
5 years	78.2 ± 5.4	90.9 ± 8.7	68.6 ± 18.6	83.3 ± 6.4	57.3 ± 15.2
10 years	71.4 ± 6.8	-	-	69.6 ± 10.4	-
Disease-specific survival; mean % ± SD					
1 year	100.0	100.0	100.0	100.0	100.0
3 years	91.6 ± 3.3	90.9 ± 8.7	100.0	94.6 ± 3.7	76.0 ± 12.3
5 years	81.6 ± 5.2	90.9 ± 8.7	68.6 ± 18.6	87.7 ± 5.8	64.7 ± 14.8
10 years	74.8 ± 6.7	-	-	74.0 ± 10.3	-

Dashes signify unavailable data.
KM, Kaplan-Meier; SD, Standard Deviation.

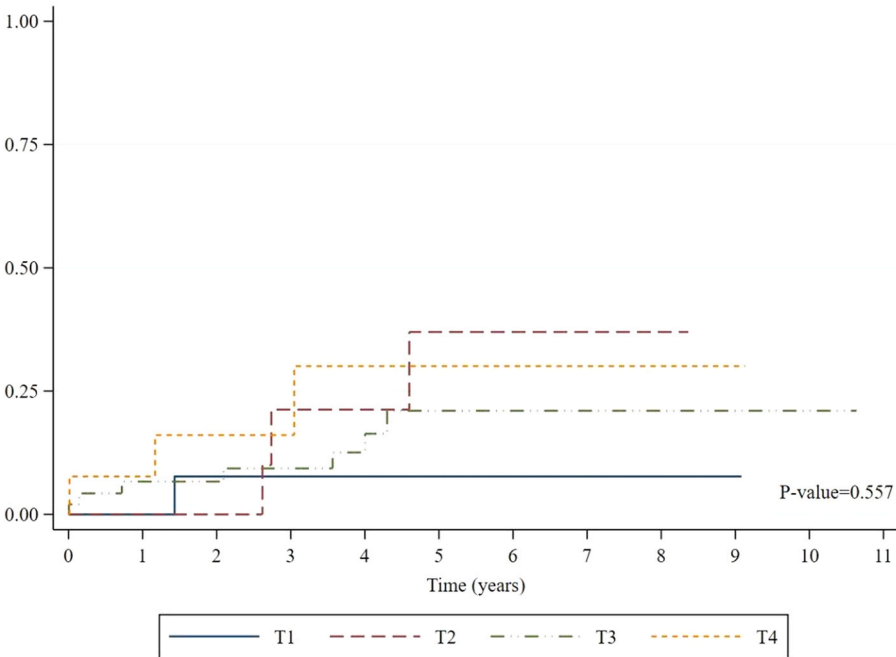


Fig. 1. Kaplan-Meier analysis of systemic metastasis stratified by T category according to the 8th edition of the AJCC staging system for uveal melanoma.

hazards model for cause-specific hazard in competing risk analysis, the incidence rate of melanoma-related death was 3.4 cases per 100 person-years. Of the 13 patients who died from melanoma-related metastasis, histopathological analysis was available for 10 cases. Among these, 2 tumors were classified as epithelioid cell type, while the remaining 8 tumors exhibited a mixed cell type.

Factors predicting survival outcomes were evaluated using Cox regression analyses. Survival was not significantly associated with the duration of symptoms ($P = 0.70$), initial visual acuity ($P = 0.91$), basal

tumor diameter ($P = 0.64$), or ciliary body involvement ($P = 0.61$). After adjusting for T stages, the hazard ratios for metastasis-related death in T2, T3, and T4 compared to T1 were 2.55, 2.10, and 6.05, respectively; however, these differences were not statistically significant. Similarly, the hazard ratios for primary treatment with brachytherapy and combined endoresection with brachytherapy, compared to primary enucleation, were 0.74 ($P = 0.66$) and 0.47 ($P = 0.48$), respectively, with no statistically significant differences observed. Univariate and multivariate analyses identified distant metastasis as the

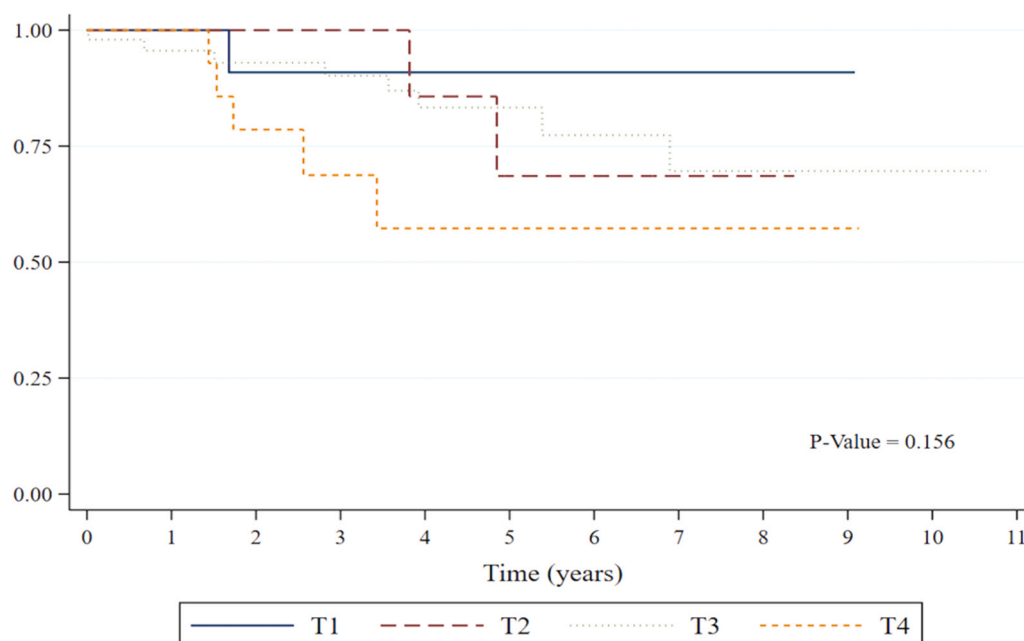


Fig. 2. Kaplan-Meier analysis of overall survival stratified by T category according to the 8th edition of the AJCC staging system for uveal melanoma.

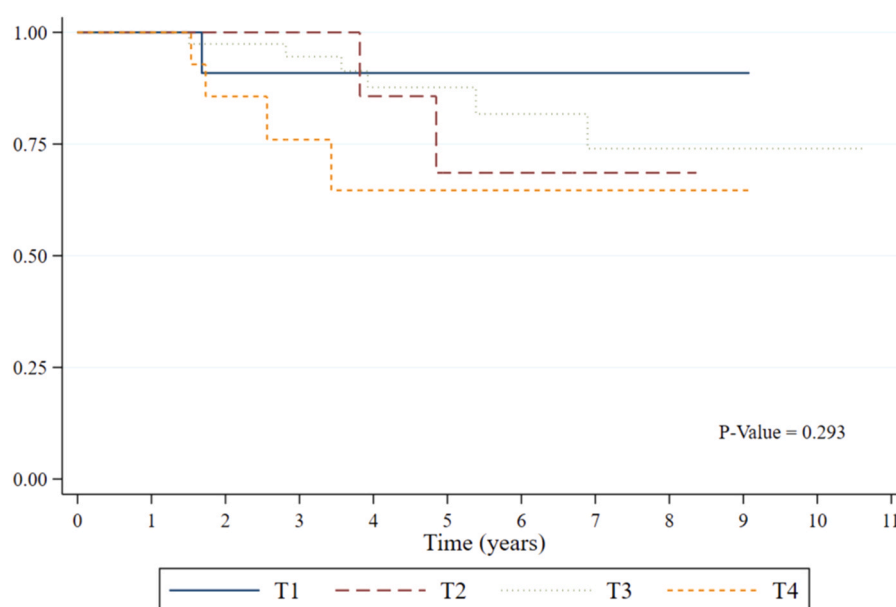


Fig. 3. Kaplan-Meier analysis of disease-specific survival using a competing risk model, stratified by T category according to the 8th edition of the AJCC staging system for uveal melanoma.

only factor significantly associated with overall survival, with a hazard ratio of 50.62 (95% CI: 10.44–245.42, $P < 0.001$).

7. Discussion

In our series, the majority of patients were Asian. The mean age at diagnosis in our cohort was 51.6 years, which is consistent with findings from other studies in Asian populations, such as Korea (51.9 years), India (47 years), Singapore (42.9 years), and a combined regional study (52 years).^{7,8,13,14} This suggests that uveal melanoma tends to present at a younger age in Asian populations compared to Caucasian populations, where the mean age at diagnosis has been reported to range between 57 and 60 years.^{5,6} Several studies from Asian populations have reported a larger mean basal tumor diameter in Asians compared to Caucasians.¹⁵

The mean largest basal diameter in various Asian studies has ranged from 12 to 14.8 mm,^{5,15–17} which is comparable to the 11.9 mm reported in our study. Although medium-sized tumors, as classified by the COMS, were the most common in our cohort, consistent with findings in both Asian and Caucasian populations, large-sized tumors also represented a significant proportion and were closely aligned with the medium-sized group.

When categorizing patients with UM using the 8th AJCC staging system, the T-staging distribution in our cohort was 19% in T1, 18% in T2, 48% in T3, and 15% in T4. Comparatively, an Indian study reported T1 (11%), T2 (23%), T3 (39%), and T4 (12%),¹⁴ while in the USA, the distribution was T1 (46%), T2 (27%), T3 (21%), and T4 (6%).¹⁸ These findings highlight a higher frequency of late-stage tumors (T3 and T4) among Asians compared to Caucasians.

The differences in tumor staging and earlier age at diagnosis among Asian populations may reflect genetic, environmental, or demographic factors. Potential contributors include variations in genetic predisposition, healthcare access, tumor biology, and healthcare-seeking behavior, which may influence disease progression and detection.

Presenting VA significantly differed between T1 and higher T stages (T3, T4), with lower T stages more likely to present with better initial VA compared to higher T stages. Regarding tumor characteristics, a dome-shaped configuration was most common in T1 tumors (77.8%), while a mushroom-shaped configuration predominated in T4 tumors (85.7%), likely due to tumor growth breaching the Bruch's membrane and expanding outward, a hallmark of advanced disease. The incidence of associated subretinal fluid in our cohort (89%) was comparable to previous studies from India (89%) and the USA (79%).¹⁴ While subretinal fluid is a hallmark of uveal melanoma, its lack of correlation with T-stage in our cohort may suggest that subretinal fluid production is driven more by tumor vascular permeability and metabolic activity than by tumor size alone. Even small (T1) tumors can present with significant fluid if they are actively growing, whereas larger, more chronic tumors may have relatively stable or localized fluid. Shields et al., in a large study, reported that based on the AJCC staging system, increasing tumor stage was associated with older age, larger tumor size, a higher incidence of subretinal fluid, and greater likelihood of extraocular extension.¹⁸

Extraocular extension, a critical prognostic factor, was reported by Shields et al. to increase progressively with T stage, with rates of 1%, < 1%, 4%, and 12% for T1, T2, T3, and T4, respectively.¹⁸ In our study, the overall incidence of extraocular extension was 1.9%, which is comparable to the rate reported in their study. However, in our cohort, extraocular extension was observed exclusively in T4 tumors. This limitation to T4 may be attributed to the smaller sample size of our study, which might not adequately represent earlier T stages.

It is also important to consider the referral patterns specific to our healthcare setting. Given the limited number of ocular oncology centers in Thailand, our institution receives referrals for suspected intraocular tumors across the entire spectrum of severity, including small lesions. Therefore, the high proportion of advanced-stage (T3 and T4) tumors observed in our cohort likely reflects delayed initial diagnosis or patient presentation, rather than a referral bias that selectively filters out early-stage cases.

The treatment approach for uveal melanoma is determined by multiple factors, including tumor stage, size, thickness, location, and patient preference. Available treatment modalities include plaque brachytherapy, charged particle irradiation, local resection, enucleation, and, in advanced cases, orbital exenteration.⁵ Among these, plaque brachytherapy and enucleation are the most commonly utilized treatments.^{5,6}

At our institution, ¹⁰⁶Ru plaque brachytherapy is available as a primary treatment modality. In accordance with the manufacturer's recommendations, ¹⁰⁶Ru brachytherapy is generally effective for tumors with an apical thickness of up to approximately 6 mm, leveraging the penetration depth of beta radiation.^{19,20} Tumors exceeding this thickness or those beyond the plaque's effective coverage are typically considered for enucleation. At the time of this study, alternative modalities such as proton beam radiotherapy were not available, making enucleation or endoresection the primary treatment options for tumors beyond the effective range of ¹⁰⁶Ru brachytherapy.

A systematic review and meta-analysis of ¹⁰⁶Ru brachytherapy reported a radiation dose to the tumor apex ranging from 70 Gy to 250 Gy, with local control rates between 59% and 98%.²¹ The overall weighted mean local control rate was 84%,²¹ which is comparable to the 83.3% primary tumor control rate observed in our study. Our findings align with these reported outcomes, reinforcing ¹⁰⁶Ru brachytherapy as an effective treatment for appropriately selected tumors. However, tumor thickness remains a critical factor influencing treatment success. Given that beta radiation penetration of ¹⁰⁶Ru is limited to approximately 6 mm, tumors exceeding this threshold are at a higher risk of incomplete

regression, necessitating additional interventions such as proton beam radiotherapy, endoresection or enucleation.

Enucleation was the primary treatment for uveal melanoma before the introduction of brachytherapy and remains a common approach, particularly for large tumors. In the Asian context, primary enucleation is the most frequently utilized treatment modality for uveal melanoma, except in China and Israel, where brachytherapy is more commonly performed.^{5,15,22} Reported rates of primary enucleation vary widely, ranging from 18% to 100%.⁵ In our series, enucleation was the most frequently performed treatment, accounting for 53.8% of cases, primarily for large tumors with a mean tumor thickness of 11.9 mm (range 5.0–18.0 mm). One patient with a 5 mm thick tumor underwent enucleation due to its epipapillary location, which was unsuitable for brachytherapy and carried a high risk of optic nerve invasion.

Final VA did not differ significantly across T-stages, suggesting that visual prognosis is primarily dictated by baseline tumor location and anatomical involvement rather than treatment modality. This finding is likely influenced by a globe salvage bias, as the high rate of primary enucleation in advanced stages (T3 and T4) restricted the subset of patients available for long-term visual acuity analysis.

Tumor cell type is recognized as a significant prognostic factor influencing survival outcomes in patients with uveal melanoma.^{23–26} The modified Callender classification has demonstrated an improved correlation between cell morphology and patient mortality.²³ Previous research has indicated that spindle cell melanomas are associated with the most favorable prognosis, while mixed cell melanomas have an intermediate prognosis, and epithelioid cell melanomas exhibit the poorest survival outcomes.^{23–26} Studies from India and China have reported a higher incidence of spindle cell tumors compared to mixed cell-type tumors.^{15,27} However, in our study, the majority of patients with available histopathological results (85.7%) had mixed cell type, and among those who died from melanoma-associated metastasis, 80% also had mixed cell type tumors. According to the AJCC T category, no statistically significant differences were observed between cell types and T classification, likely due to the disproportionate distribution of cases among the different cell type groups.

Shields et al. reported a large cohort of 8100 patients with uveal melanoma, comprising 7918 (98%) Caucasians, 105 (1%) Hispanics, 44 (<1%) Asians, and 33 (<1%) African Americans.⁶ In terms of prognosis, the 10-year metastatic rates were 25% in Caucasians, 12% in Hispanics, 36% in Asians, and 8% in African Americans.⁶ However, studies specifically evaluating the prognosis of uveal melanoma in non-Caucasian populations, particularly Asian patients, remain limited. According to the AJCC T category, Kaplan-Meier estimates of metastasis at 5 years in our study were 8% for T1, 37% for T2, 21% for T3, and 30% for T4. In comparison, Shields et al. reported metastatic rates of 8% for T1, 14% for T2, 31% for T3, and 51% for T4.¹⁸ While our cohort demonstrated a higher metastasis rate for T2 tumors, it is important to note that T3 was the most common classification in our study, with fewer cases in the T2 and T4 categories.

Furthermore, these discrepancies should be interpreted in the context of follow-up duration. The Shields et al. study represents a significantly larger cohort with extended long-term monitoring. In contrast, our median follow-up of 37 months implies that our 5-year Kaplan-Meier estimates are derived from a smaller number of patients remaining at risk. This shorter observation period may limit the stability of our estimates, potentially contributing to the higher variability observed in the T2 group and the lower-than-expected rates in the T4 group compared to Western benchmarks.

Liu et al. reported a series of 583 uveal melanoma patients in China, where 78% of tumors were classified as medium-sized. The most employed treatment modalities were brachytherapy (71.3%) and primary enucleation (20.4%). Metastatic disease, detected through general examination, was identified in 2.9% of cases, while 80.6% of patients were alive without evidence of metastases or tumor recurrence at a median follow-up of 30 months. However, follow-up data were

unavailable for 11.0% of patients.¹⁵ Compared to our cohort, where 50% of cases were medium-sized tumors, the median follow-up duration was longer at 37 months. At initial presentation, two patients (1.9%) had distant metastases, while an additional 12 patients (11.5%) developed distant metastases during follow-up. The slightly higher metastasis rate in our study may be attributed to larger tumor sizes and a longer follow-up period. This is consistent with previous studies demonstrating that each millimeter increase in tumor thickness is associated with an approximately 5% increase in the risk of metastasis over 10 years, with a hazard ratio of 1.08.² Larger tumor size is a well-established prognostic factor for metastatic risk, highlighting the importance of early detection and timely intervention in improving patient outcomes.

In our series, the overall survival rates at 1, 3, 5, and 10 years were 98%, 88%, 78%, and 71%, respectively, which were slightly lower than those reported by Liu et al., where the 5- and 10-year survival rates were 92.7% and 85.1%, respectively.¹⁵ The incidence rate of overall mortality in our study was 4.2 cases per 100 person-years. Mortality rates stratified by T category demonstrated an increasing trend with tumor stage, with rates of 1.4, 3.5, 4.1, and 9.6 cases per 100 person-years for T1, T2, T3, and T4, respectively, reflecting the progressive nature of advanced disease. The only significant predictor of survival outcome in both univariate and multivariate analyses was the presence of distant metastasis, with a hazard ratio (HR) of 50.62 (95% CI: 10.44–245.42, $P < 0.001$). Kaplan-Meier estimates of disease-specific survival at 5 years were 91% for T1, 69% for T2, 88% for T3, and 64% for T4 in our cohort. In comparison, the 5-year survival rates reported by Shields et al. were 96% for T1, 92% for T2, 81% for T3, and 70% for T4.¹⁸ Their study also demonstrated that the risk of metastasis and death doubled with each increasing melanoma category. These findings reinforce the strong correlation between tumor stage and survival, emphasizing the need for early detection and aggressive management in advanced cases to improve long-term outcomes.

It is important to emphasize that treatment selection in our cohort was non-randomized and largely determined by baseline tumor size, stage, and anatomical location. Consequently, the comparisons of survival and globe salvage outcomes across modalities are descriptive in nature and should be interpreted with caution. These findings reflect clinical practice patterns rather than evidence of superiority or equivalence between treatment approaches.

Our study has several limitations. First, although our median follow-up of 37.3 months is comparable to other Asian series, it is relatively short given the known risk of late metastasis in uveal melanoma. This duration limits the assessment of long-term outcomes; consequently, the 10-year metastasis and survival estimates should be interpreted with caution, as they are projected from a limited number of patients remaining at risk during this extended interval. This may lead to an underestimation of late events compared to Western series with decade-long monitoring. Second, genetic and molecular profiling were not available for our cohort. In our current practice setting, molecular testing is not reimbursed by the government healthcare system and is therefore not routinely performed, even for enucleated specimens. This precluded the inclusion of molecular classification in our prognostic analysis.

8. Conclusion

This study represents the largest report on uveal melanoma in the Southeast Asian region, providing comprehensive data on the clinical and pathological characteristics, treatment modalities, outcomes, and prognosis of uveal melanoma patients. Although the incidence of uveal melanoma in Asians is lower than that reported in Caucasian populations, our findings indicate that Asian patients present at a younger age, with larger mean basal tumor diameters and greater tumor thickness at diagnosis. Clinical presentation remains straightforward, with enucleation and plaque radiotherapy being the most commonly utilized treatment modalities. Tumor staging, based on the 8th AJCC

classification, was significantly associated with survival outcomes, with advanced T stages correlating with lower survival rates. Additionally, distant metastasis was identified as a key prognostic factor, significantly impacting overall survival. These findings highlight the need for early detection, optimized treatment strategies, and long-term follow-up to improve outcomes in Asian patients with uveal melanoma.

CRediT authorship contribution statement

CRediT

Funding

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